From Material Turnover to Causal Memory: A Falsification Ladder for Organizational Continuity in a Simulated Droplet

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Abstract

Biological individuals persist while their constituent material is replaced. We ask a deflationary, testable version of this question in a simulated reaction-transport droplet: can an internal state encode the droplet's experience in a way that is causal, transplantable, and robust to constituent turnover — and does such memory individuate droplets? Using a droplet with a designed two-component intensive memory field coupled to nutrient uptake (m+) and attractant production (m-), we run a preregistered ladder of interventions with grouped, prospective, kill-switched evaluation and continuity-safe entity tracking. **Central result:** under a globally imposed nutrient history, multiple persistent droplets locally encode a similar engineered cumulative-experience coordinate (h1); a continuously tracked droplet retains this internal causal state during substantial constituent turnover (longitudinal holdout on untouched seeds: h1 held-out $R2=0.98$, 95% CI [0.97,1.00] at old-material fraction M~0.19, with 36/36 track survival). h1 is causally expressed through uptake and transplantable into a fresh standardized body ($R2=0.61$), and exceeds body-size/mass baselines after turnover. Crucially, h1 is *global* in informational content — decodable from essentially any large droplet — so it does **not** individuate droplets. A second candidate coordinate (h2; temporal order / value dispersion) is decodable at rest but fails to qualify: causal response ~1-D, and deep-turnover held-out $R2=0.34$ (<0.50) under continuity-safe tracking. Methodologically, we show that high internal variation, decodability, rest-state causal expression, and largest-component visual continuity each fail to imply a turnover-surviving, individuating causal memory, and that a largest-component *reselection* tracker produces spurious continuity that a longitudinal tracker removes. We claim no life, agency, identity, individuation, reproduction, or heredity.

1. Introduction

Persistence of *organization* rather than of *matter* is a candidate substrate for biological memory and identity [7, 8]. A droplet whose molecules turn over but whose organization endures might carry a history of its experience. We make this precise and falsifiable. A history variable being *decodable* from a system is insufficient evidence of memory: a decoder may exploit residual environment, current body size, laboratory-frame position, or the mere fact that a globally imposed history is written into every entity. A memory worth the name must (i) be causal — it must shape dynamics [9]; (ii) be separable from the current body and environment, e.g. survive transplantation into a standardized body; and (iii) survive substantial replacement of the constituent material. Individuation — making one droplet distinguishable from another by its history — requires strictly more than persistence and is not claimed here.

2. Related work

Our droplet is a dissipative, pattern-forming reaction-transport system in the tradition of reaction-diffusion morphogenesis [1, 5, 6], with excitable local chemistry of FitzHugh–Nagumo type [2, 3]. The memory field is an engineered attractor-like internal state; distributed attractor memories are classical [4]. Unlike associative-memory networks, our aim is not storage capacity but the *causal and material* status of a history coordinate under turnover, evaluated with the prospective, preregistered, kill-switched discipline now standard in empirical model evaluation.

3. Model

The substrate is a periodic 64x64 lattice (dt=0.1) carrying a density field rho, internal excitable chemistry (U,V), nutrient N, and attractant c, with advective transport toward c, bounded growth, and death. On top of the frozen scaffold we add a bounded intensive memory m=(m1,m2) (extensive Mf=rho m), written by a local experience signal and read through two existing physical channels.

Write. The experience signal is Psi=tanh(k_{exp}(N-c)+k_up(uptake-mean_uptake)). Each component is a leaky integrator of Psi with its own time constant, plus local templating and diffusion:

> dm_k/dt = eta_w*Psi - eta_d,k*m_k + eta_t*(_nbr - m_k) + D_m*lap(m_k), with m_k in [-1,1].

New mass created by growth inherits the local intensive memory (Mf mathrel{+}= g m); death scales all fields uniformly. **Read.** m+=m1+m2 modulates nutrient uptake (uptake->uptake (1+lambda+tanh m+)) and m-=m1-m2 modulates attractant production (propto 1+lambda-tanh m-). With lambda-=0 the model reduces bit-identically to the single-channel predecessor; with all coupling zero it reduces to the frozen scaffold. **History coordinates.** Two-phase *global* nutrient histories (Nmathrel{+}=a during each phase) define h1=a_early+a_late (cumulative) and h2=a_late-a_early (order).

Table 1 — Frozen C1c parameters.

symbol	value	meaning
eta_w	0.015	experience write gain
eta_d1,eta_d2	0.35, 0.006	fast / slow forgetting
eta_t	0.010	local templating (4-neighbour)
D_m	0.010	memory diffusion
k_{exp},k_up	1.0, 1.0	experience-signal gains
lambda+,lambda-	0.25, 0.15	uptake / attractant coupling
grid, dt	64x64, 0.1	lattice, timestep

4. Methods

Protocol. Warm a body (2000 steps), erase memory, apply the two-phase history (T=60 each), settle 20 steps, then relabel all material "old" (cohort 0) and evolve forward; new growth is cohort 1. The old-material fraction M is the old cohort's mass share of the tracked entity; the pulse-chase tracer is passive and never enters the dynamics. **Entity detection and tracking.** The substrate self-organises into many small droplets. Connected components are detected periodically (wrapped labelling; circular centroids); feature extraction is translation-invariant (verified: whole-world shifts, including boundary-straddling, leave size and memory statistics bit-invariant). Historical analyses selected the *largest* component — but this is *reselected* each checkpoint (§5.9); we add a frozen **longitudinal tracker** (initialise once; match by periodic maximum mask-overlap; censor on loss; never uses h1, h2, M, or labels) and re-certify all turnover claims longitudinally. **Decoding and statistics.** Grouped leave-history-out ridge regression (rows sharing a history stay in one fold; no row-wise leave-one-out); development and prospective families separated; sealed families hashed before selection and executed once; donor-level bootstrap 95% CIs; constant, shuffled-history, and trivial-feature nulls.

5. Results

5.1 Persistent organization; hidden state is a generic attractor. The droplet remains localized and viable under turnover; a hidden internal state exists but behaves as a generic causal attractor, not a history-specific identity (individuation AUC ~0.3–0.4). **5.2 An engineered causal memory h1.** The two-component field is causal and erasable (ablating coupling removes the effect exactly), transplantable, and turnover-related. **5.3 Temporal order made readable.** A second orthogonal read-out (m–> attractant) makes matched-dose temporal order causally readable; the order contrast collapses sim70x/73x (dev/prospective) under lambda- ablation and is absent on the uptake axis (Fig 4). **5.4 Apparent dimensionality corrected.** The original "high-dimensional" continuous-history claim was an artefact of a 20–47x history-amplitude mismatch and a replicate-leaking row-wise cross-validation; grouped evaluation reduces the headline R2 from 0.57 to 0.19; three estimators agree the frozen writing is rank ~1 (Fig 5). **5.5 Two-coordinate storage at rest.** A minimal de-saturating writing change (C1c) yields two prospectively decodable coordinates at rest (h1,h2~0.94); the rank ratio is 0.283, marginally below the preregistered 0.30. **5.6 Causal-response collapse.** Despite 2-D storage, the causal response is ~1-D: h1 is causally expressed (transplant 0.61, in-place 0.50) but h2 is not (-0.04/0.30) (Fig 5). **5.7–5.8 h2 is a frame-free dispersion; deep-turnover kill-switch.** h2 is carried by a frame-free value-dispersion statistic (lab-frame decoder collapses under translation), with family-dependent turnover behaviour; on a sealed family executed once, held-out h2~0 at M~0.21 (Fig 6). **5.9 Tracker-continuity audit (Fig 7 correction).** Human inspection of the viewer exposed the tracked-entity overlay teleporting between blobs. An event audit found the selection rule (argmax size, reselected each frame) switches entities 5 times on the incident seed, with M and text{std}(m-) discontinuities at those frames. Detection is periodic-safe, so this is *reselection*, not a wrap artefact. A frozen longitudinal tracker on **untouched** seeds 38502–38504 eliminates switching (0 switches, 36/36 survival) and re-certifies h1: init 0.92 [0.86,0.99]; deep 0.98 [0.97,1.00] at M~0.19; h2 stays 0.34 (<0.50). h1 decodes ~equally from largest/2nd-largest/longitudinal (0.96/0.98/0.99) because h1 is *global.* **5.10 Surviving result.** h1 is a one-dimensional, global, causal, transplantable memory of cumulative experience, retained by a continuously tracked droplet through substantial turnover; it does not individuate droplets.

6. Discussion

The ladder separates concepts usually conflated: memory vs decodability; storage vs causal expression; passive local copying vs active reconstruction; material replacement vs organizational inheritance; and visual/size-reselected continuity vs longitudinal continuity. h1 storage and causal expression are *local*, but its informational content is *global*, so it does not individuate. Individuation would additionally require multiple independent, causal, transplantable, turnover-surviving coordinates

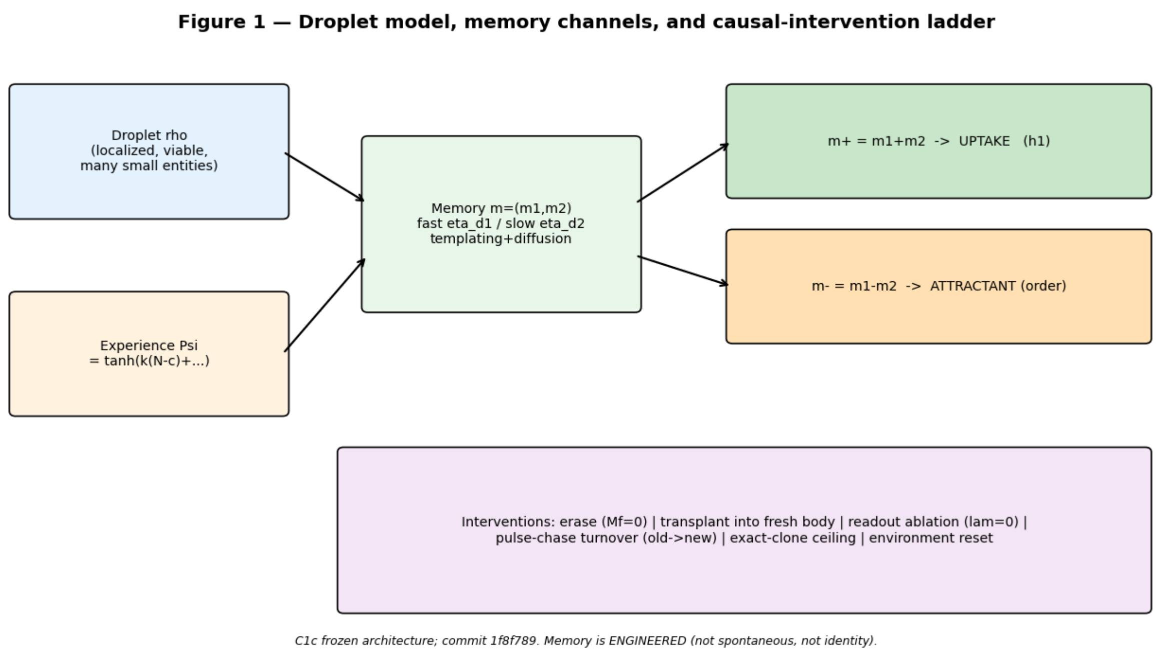


Figure 1: Model and causal-intervention ladder (schematic). Frozen C1c architecture: write Psi, two-timescale memory m=(m1,m2), read-outs m+->uptake, m–>attractant, and the intervention suite. Source: docs/paper/fig1_model_ladder.png.

and distinct histories carried by different droplets in one world — none demonstrated. The tracker incident, handled transparently, converts a loose "the entity survived turnover" into a certified "a continuously tracked droplet retains a global cumulative coordinate through turnover, which does not individuate."

7. Limitations

Engineered (not emergent) memory; simulator-specific dynamics; small entities (sim30–55 cells); designed two-phase *global* histories; low-dimensional, global, non-individuating h1; explicit local memory copying during growth; historical analyses used largest-component reselection (corrected here); mean-field transplant read-out; no reproduction; no active reconstruction; no individuation; substrate-general capacity untested.

8. Conclusion

Under a globally imposed history, persistent droplets locally encode a similar engineered one-dimensional cumulative causal memory; a continuously tracked droplet retains this internal causal state through substantial material turnover. It does not individuate droplets, and a second decodable coordinate did not qualify as an independent, causal, turnover-surviving dimension. Individuation, reproduction, and a genome are not warranted by the evidence.

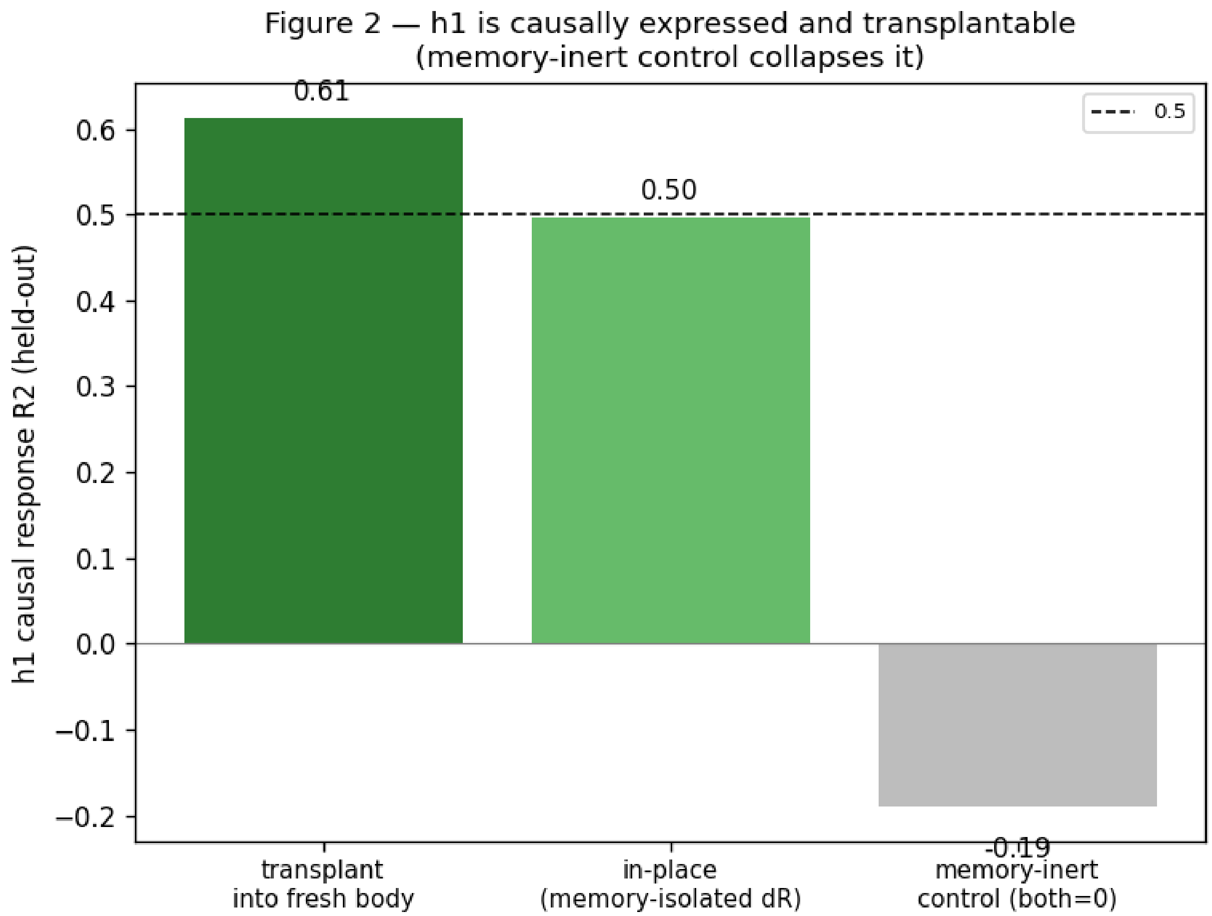


Figure 2: h1 causal memory. Held-out h1 response after mean-transplant into a fresh standardized body (0.61), in-place memory-isolated (0.50), and the memory-inert control (collapse), n=12 histories, prospective. Data: results/wd01_phasec/phasec_causal_*_raw.pkl.

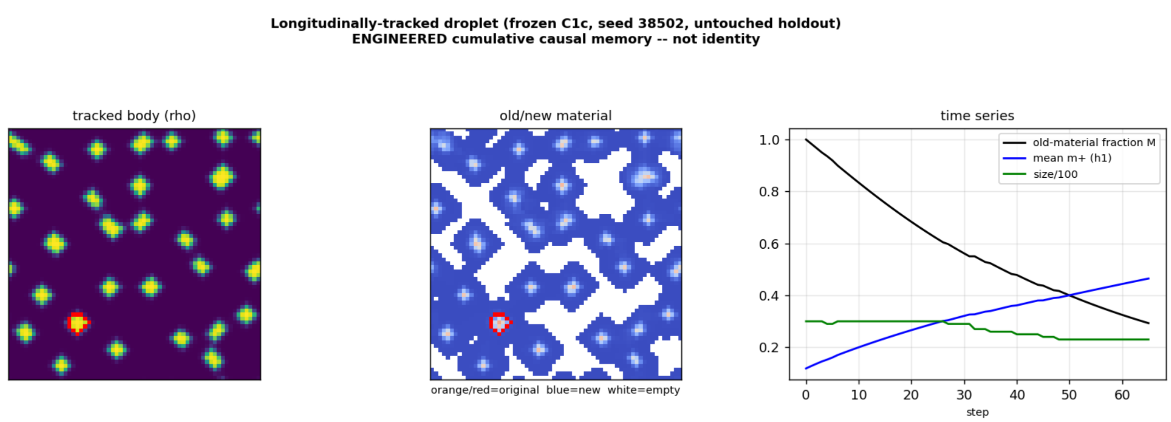
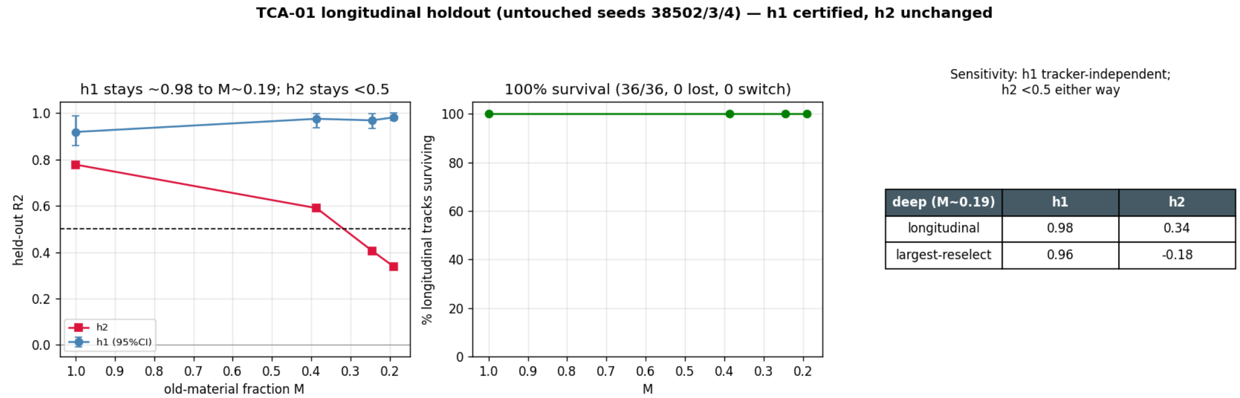


Figure 3: Longitudinally-tracked turnover (frozen C1c, untouched seed 38502). Tracked body and old/new material carry the same tracked-entity contour; the time series shows old-material fraction M declining, mean m+ (h1) rising, and stable size. Source: docs/paper/figures/tracked_3panel_en.png; scripts/observe_droplet.py.



deep (M~0.19)	h1	h2
longitudinal	0.98	0.34
largest-reselect	0.96	-0.18

Figure 4: Longitudinal holdout certification (untouched seeds 38502–38504, n=12 histories x3 seeds, executed once). h1 held-out R2 with 95% CI vs M; 100% track survival; sensitivity table (largest vs longitudinal). Data: results/observer/tca_holdout_raw.pkl.

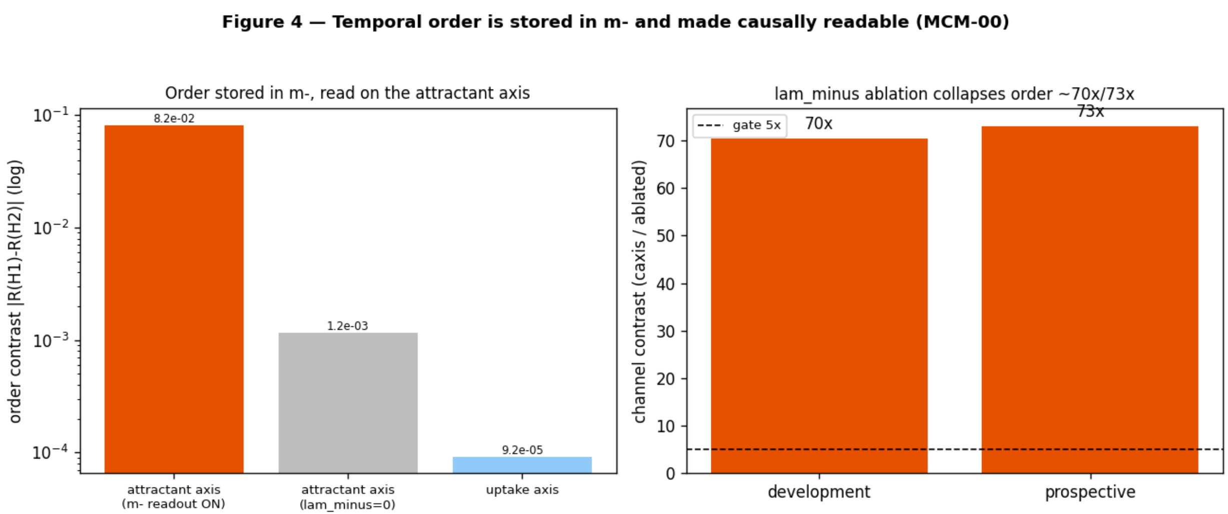


Figure 5: Temporal order stored in m- and made causally readable (MCM). Order contrast on the attractant axis vs lambda–ablated vs uptake axis, and the sim70x/73x channel contrast (dev/prospective). Data: results/sc_mcm/certificate.json.

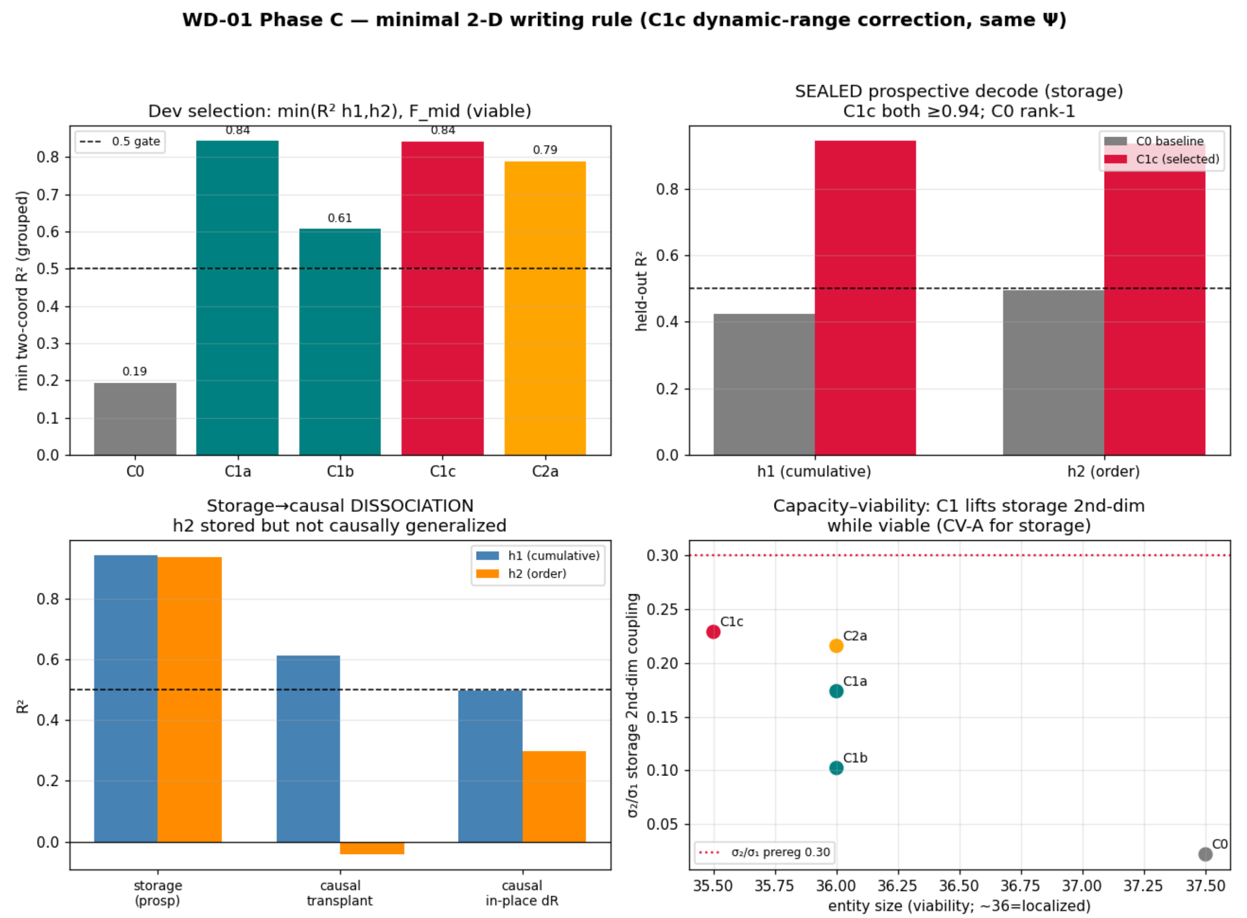


Figure 6: Storage vs causal dimensionality (WD-01 Phase C). Two-coordinate storage at rest with ~1-D causal response; grouped-evaluation correction. Data: results/wd01_phasec/*_raw.pkl.

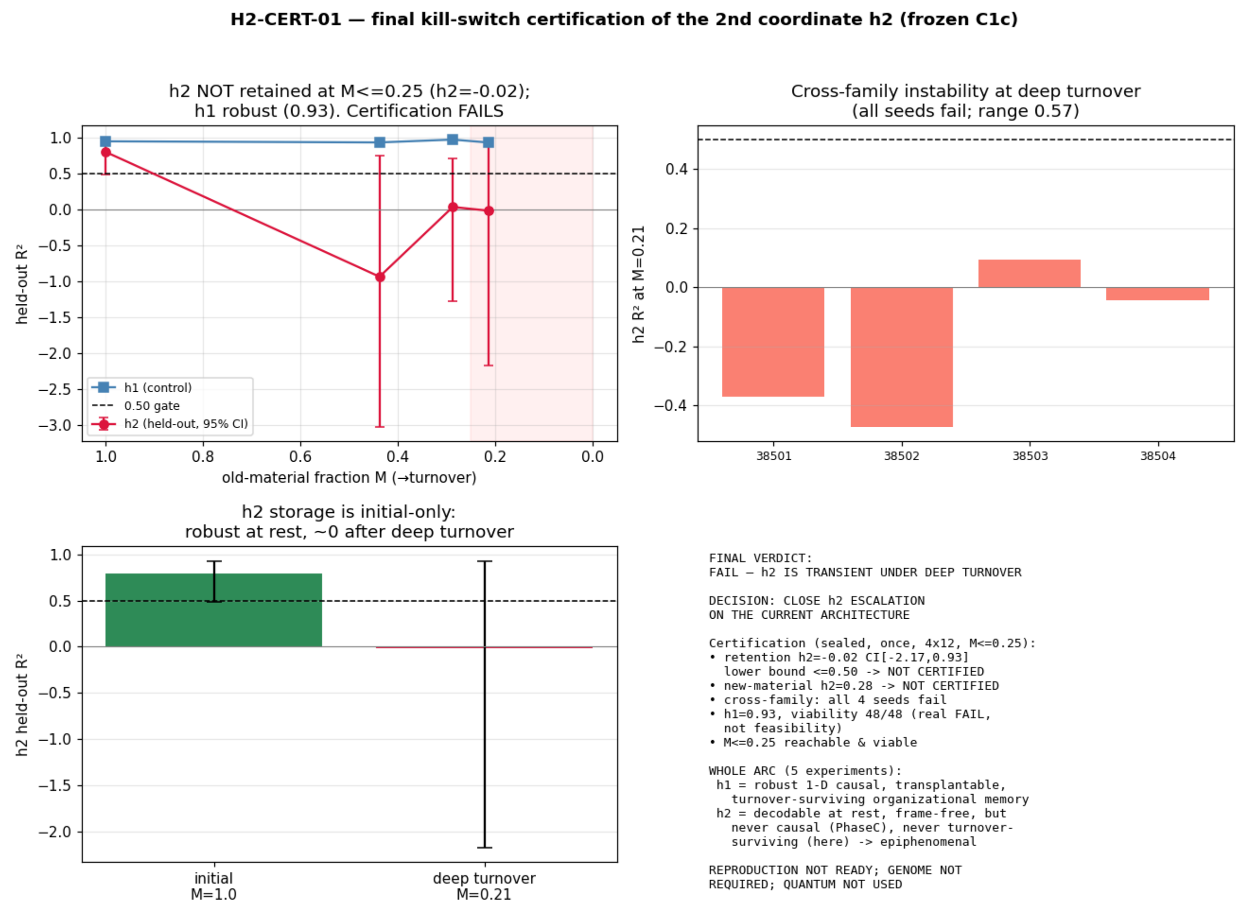
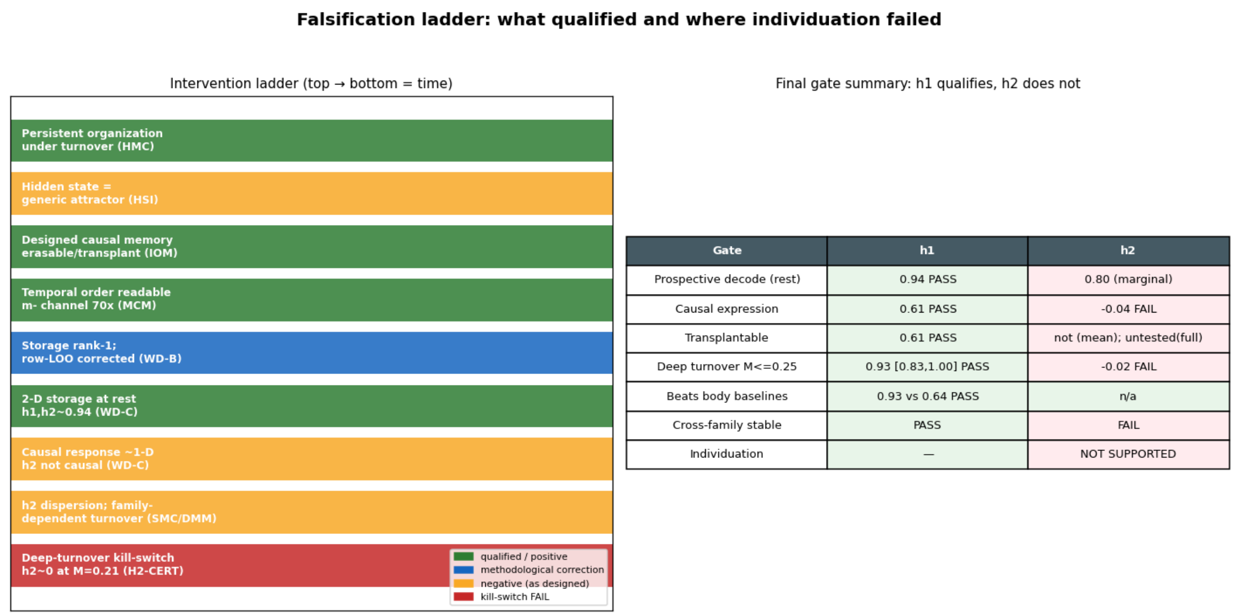


Figure 7: h2 deep-turnover kill-switch (H2-CERT). Sealed held-out h2~0 at M~0.21 with 48/48 viability; family-dependence. Data: results/h2cert/h2cert_sealed_raw.pkl.



Gate	h1	h2
Prospective decode (rest)	0.94 PASS	0.80 (marginal)
Causal expression	0.61 PASS	-0.04 FAIL
Transplantable	0.61 PASS	not (mean); untested(full)
Deep turnover M<=0.25	0.93 [0.83,1.00] PASS	-0.02 FAIL
Beats body baselines	0.93 vs 0.64 PASS	n/a
Cross-family stable	PASS	FAIL
Individuation	—	NOT SUPPORTED

Figure 8: Falsification ladder and final gate summary: h1 qualifies, h2 does not; individuation fails. Source: docs/paper/paper_claim_ladder.png.

Data and code availability

All code, frozen configurations, sealed manifests (SHA-256), raw data, analysis scripts, and figure generators are in the repository under isolated branches (see Reproducibility appendix and docs/paper/REPRODUCIBILITY.md). No data were withheld; negative results and corrections are retained.

Reproducibility statement

Physics is frozen (engine blob 7c91b91; C1c parameters in Table 1). Sealed prospective families were hashed before selection and executed once. Grouped leave-history-out decoding avoids replicate leakage. The longitudinal tracker is frozen and specified in docs/audit/TCA_01_TRACKER_SPEC.md. Reproduction commands are in the Reproducibility appendix.

References

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